Should the first-principles agent file be split

First-Principles Analysis

10 October 2026

Professional Strategic Analysis

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Read with: How to read this analysis · All files

Executive Summary

Recommendation: Do not split the file first. Extract the mechanical publishing pipeline into a script/hook, then re-measure the generated artifact's real token count and redefine the budget around that measurement, then decide any further companion-technique splitting on real firing-frequency evidence (chain C3).

Band (from §6): MEDIUM — matches chain C3's own band, the weakest chain the Conclusion rests on (chain C3).

Would change it: Having the trade-off's weights set by whoever bears the implementation cost, and real measurement of the generated artifact's token count or of how often each companion technique actually fires (chain C3).

1. Problem Essence

Core problem: Given that a single agent-definition file has grown far past a recorded line/token budget, no structural fix (splitting the file, retiring the budget, or narrowing the agent's responsibilities) can be chosen correctly until it is known *what real cost or quality property the budget was meant to protect* and *whether the proposed fix actually changes that property* rather than merely changing the number the gate reads.

Success criteria:

1. The analysis names, with cited evidence, which candidate goal(s) the budget plausibly protects, or states precisely what evidence is missing to decide — a reader can check this against the Ground Truths section without further clarification.
2. The analysis tests the "split the file" intervention against how content is actually loaded by the harness (always-loaded vs. fetched-on-demand vs. never-loaded-unless-invoked), not against the file's line count alone — a reader can verify this by checking whether each option's score traces to a loading-mechanics ground truth.
3. The analysis produces a decision among split-only, retire/redefine-the-budget-only, narrow-scope-only, or a hybrid/composite of these, selected by a transparent weighted comparison rather than by default — a reader can recompute the comparison's arithmetic and flip test independently.
4. The analysis names any residual open question whose answer would change the recommendation, and states exactly what evidence would resolve it — a reader can check this against the stated open questions in the Conclusion section.

2. Assumptions Table

Type	Accept	Challenge	Discard
physical law	2	0	0
current constraint	1	0	0
convention	0	3	0
untested belief	0	11	2

Shading darkens with count.
An untyped (legacy) row appears only for assumptions with no type.

Companion techniques applied in this phase: Inversion (inverting "splitting fixes the problem" into "splitting guarantees nothing is fixed," enumerating the preconditions that would have to hold for a split to be a real fix — these became A-3, A-4, A-5, A-7 below) and Fishbone (breadth-first brainstorm of *why* the file grew past budget, using the default six-category set: People, Process, Technology and Tools, Environment, Information, Resources — People: more contributors added content with no counterweight removing any; Process: no CI gate currently enforces the number (GT-8), so nothing resists accretion; Technology and Tools: a build pipeline (sync-content.py, GT-6) made adding content to 16 generated surfaces cheap and safe, removing the friction that might otherwise have limited growth; Environment: the project's entry points multiplied from 1 agent to 16 surfaces (GT-13), each drawing on the shared source; Information: the methodology itself deepened — confidence calibration, the self-audit gate, the closure ledger — which is legitimate rigor, not creep; Resources: a document-publishing capability (figures, PDF, folder index) was added, requiring new mechanical instructions (GT-11), which is the category most directly implicated in genuine scope growth. Discriminating observation: growth concentrated in "Information" is depth of the same job; growth concentrated in "Resources" is a new job. GT-11 and GT-4 below let this be checked quantitatively rather than left as impression.)

Assumption	Type	Treatment	Verdict	Verification
A-1: The original ~500-line budget was deliberately derived from a measured cost/latency target.	untested belief	Verify or flag unverified.	Challenge — no CI config, design note, or git history for this exists in the one real instance inspected (GT-8, GT-9); cannot confirm derivation either way.	unverified — flagged
A-2: The budget gate is a live CI check today, either failing or silently bypassed/raised.	untested belief	Verify or flag unverified.	Challenge — the user's own framing already treats this as open; the one real instance inspected has no CI config matching this description at all (GT-8).	unverified — flagged
A-3: If the harness must fetch a reference file on literally every invocation, splitting that content out does not reduce tokens actually consumed, and may increase it.	physical law	Accept as ground-truth candidate (this is an architecture-level invariant of token-based agent harnesses, not a current, revisable policy).	Accept — self-observed in this run: GT-1, GT-2, GT-14?	GT-1, GT-2, GT-14? (direct self-observation this run)
A-4: Content bundled with rarely-needed material inflates "always-loaded" cost regardless of in-file organization, unless the harness or a build step keeps the rarely-needed part out of the loaded artifact.	physical law	Accept as ground-truth candidate (an architecture-level invariant, not a current, revisable policy).	Accept — tautological given A-9/GT-6; corroborated by GT-6′s generator producing differently-sized artifacts from one source.	GT-6
A-5: Line count in one physical markdown file reliably proxies the agent's always-loaded token cost.	convention	Challenge before use.	Challenge — in the one real instance inspected, the always-loaded cost is a property of the *generated* artifact (GT-6), not the hand-maintained source's line count; a gate reading the wrong artifact cannot see what it claims to measure.	source: GT-6, GT-3

Assumption	Type	Treatment	Verdict	Verification
A-6: The companion-tool procedures are uniformly "auxiliary" and interchangeable candidates for lazy-loading.	untested belief	Verify or flag unverified.	Discard — as a uniform claim, GT-4 and GT-10 show the inspected system already treats individual techniques differently (all eight inlined, but two of them additionally forced through a fresh Read at point of use), which only makes sense if frequency/fidelity differ technique-by-technique.	GT-4, GT-10
A-7: Splitting a file into separate reference docs is itself what "reduces scope."	convention	Challenge before use.	Challenge — packaging (which file holds instructions) and scope (what the agent is responsible for doing) are independent variables; moving bytes to another file changes neither.	reasoning only, no external source — analyst judgment
A-8: The growth from budget to current size reflects uniform depth-of-the-same-job rather than new, distinct jobs being added.	untested belief	Verify or flag unverified.	Discard — GT-11 shows ~20% of the file's current length is document-publishing mechanics, a categorically different job from reasoning-from-ground-truths.	GT-11
A-9: This harness loads an agent definition's full body unconditionally at invocation; loads a plain reference file only when a Read-equivalent tool call fetches it; and loads a disable-model-invocation Skill's body only on that skill's own distinct explicit invocation.	current constraint	Record the expiry condition: holds only as long as Claude Code's agent/skill-loading mechanism works this way; would lapse if that mechanism changed.	Accept — self-observed (GT-1, GT-2) and corroborated by GT-5's `disable-model-invocation: true` frontmatter and the plugin's own agent/skill separation (GT-13).	GT-1, GT-2, GT-5, GT-13
A-10: A mechanical, deterministic, non-judgment procedure is better instructed as prose the model executes step-by-step than as a script a tool/hook simply runs.	convention	Challenge before use.	Challenge — the project's own precedent (GT-12: delivery-guard.py) already made the opposite choice, successfully, for a sibling concern (enforcing delivery); used as a premise in chain C2, flagged `[Assumes: A-10]` there since no script has actually been built and tested in this analysis.	GT-12; unverified for the *untested* extraction specifically — flagged
A-11: The budget optimizes for minimizing the agent's base system-prompt	untested belief	Verify or flag unverified.	Challenge — plausible, consistent with available evidence, not confirmable as *the* goal	unverified — flagged

Assumption	Type	Treatment	Verdict	Verification
token cost (candidate goal 1).			absent A-1′s missing provenance.	
A-12: The budget optimizes specifically for keeping *always-loaded* context small, as distinct from total content available (candidate goal 2).	untested belief	Verify or flag unverified.	Challenge — same evidentiary gap as A-11; if true, correctness of "split" depends entirely on A-3/A-9 holding for the specific destination chosen.	unverified — flagged
A-13: The budget optimizes for human readability/ maintainability of a single file (candidate goal 3).	untested belief	Verify or flag unverified.	Challenge — GT-6′s generator is the standard counter-move to this goal (one edited source, many generated artifacts); if this goal is real, it argues for strengthening the generator, not for manually splitting the generated output.	unverified — flagged; GT-6 bears on the counter-argument
A-14: The budget optimizes for preventing prompt bloat that degrades the agent's own instruction-following (candidate goal 4).	untested belief	Verify or flag unverified.	Challenge — partially corroborated: GT-10′s explicit rationale for forcing fresh Reads of pre-mortem/inversion ("recollection… drops the past-tense framing") is an attention/fidelity argument the maintainers already act on, suggesting something like this goal is already taken seriously in part of the system.	GT-10 (partial corroboration only)
A-15: The budget is a proxy for scope creep — an early-warning signal that the agent is doing too many distinct jobs (candidate goal 5).	untested belief	Verify or flag unverified.	Challenge, but best-supported of the five candidate goals in the one real instance inspected — GT-11 is concrete, independently identifiable added responsibility (document publishing) inside the grown file, which a pure token-cost or attention goal would not specifically predict.	GT-11
A-16: The six weighted trade-off criteria (cost-reduction, gaming-resistance, scope-creep-addressed, implementation cost, maintainability, reversibility) and their locked weights (5, 5, 4, 2, 3, 2) correctly represent the decision-relevant factors for this choice.	untested belief	Verify or flag unverified.	Challenge — analyst-set, not derived from a ground truth; load-bearing for chain C3, which is why C3 carries a MEDIUM confidence line naming this assumption. The O4-over-O3 ranking is weight-invariant (O4 dominates O3 on every single criterion, so no reweighting flips that ordering), which	unverified — flagged; partially mitigated by the dominance result in chain C3

Assumption	Type	Treatment	Verdict	Verification
			bounds — but does not eliminate — the risk this assumption is wrong.	

| A-17: A future contributor/reviewer will actually apply an “is this a script’s job or a reasoning job” test once scope is narrowed, rather than reverting to a line-count-only review. | untested belief | Verify or flag unverified. | Challenge — surfaced by the Assumption Audit on chain C3′s 2nd-order step; no review-process change has been adopted yet, only proposed. | unverified — flagged | | A-18: Some maintainer of this file lacks the ability or willingness to edit/run a script, so moving mechanical content out of markdown into code imposes a real accessibility cost. | untested belief | Verify or flag unverified. | Challenge — surfaced by the Assumption Audit on chain C3′s 2nd-order step; plausible but not confirmed for this specific project’s maintainer population. | unverified — flagged | | A-19: An open-ended commitment without a dated deliverable is systematically less likely to be completed than a dated one. | untested belief | Verify or flag unverified. | Challenge — surfaced by the Assumption Audit on chain C3′s 3rd-order step; a general project-management heuristic, not verified against this project’s own track record (GT-8/GT-9 show the *original* budget commitment’s fate is itself undocumented, which is consistent with but does not prove the heuristic). | unverified — flagged; GT-8, GT-9 (consistent with, not proof of) |

3. Ground Truths

Irreducibility check (5-Whys, reduce-to-primitives mode) on GT-11 before it is admitted: The compound claim “lines 348-630 of `agents/first-principles.md` are document-publishing mechanics, not reasoning method” reduces to: (a) is the content literally shell/typst/pandoc command text? — yes, verified by direct quotation (step headers “1. Create the file… with Bash”, “7. Draw the report figures… typst compile”, “8. Render the PDF reader report… pandoc –pdf-engine=typst”); (b) does it require judgment calls about ground truths, assumptions, or confidence, or is it deterministic given its inputs? — deterministic: every step is a fixed shell recipe with no “decide whether X constitutes evidence” branching of the kind Phases 1-5 require elsewhere in the same file. Both primitives bottom out at direct textual observation (a definition/measurement, not an inference), so GT-11 passes the irreducibility test and is admitted below without a `?`.

GT-1 This analysis’s own context, at the start of this run, already contained the full text of the agent’s methodology including all 8 inlined companion-technique procedures (five-whys, fishbone, inversion, pre-mortem, trade-off, second-order, estimate, theoretical-limit) with no Read tool call issued for any of it — source: direct inspection of this run’s own system prompt/context; read-at-source: the companion-technique “## Procedure” sections, present verbatim in this run’s instructions from the first turn.

GT-2 `output-template.md` and `validation-rubric.md` were absent from this run’s initial context; each was fetched during this run via an explicit Read tool call, matching the methodology’s own instruction (“Open the output template once, before the first section is appended… Use Read…”; “Open the Self-Audit Gate’s rubric once… Use Read…”) — source: this run’s own tool-call transcript (the two Read calls issued earlier in this conversation) plus direct quotation of the instruction text; read-at-source: the quoted instruction passages, and the fact that this analysis could not quote either document’s content before issuing those two Read calls.

GT-3 In the one real instance inspected (the `first-principles` plugin repository), `agents/first-principles.md` is 1,390 lines and 133,214 bytes (≈33,300 tokens at a 4-chars/token approximation) — already beyond both the ~500-line budget and the ~878-line “two milestones later” figure named in the scenario — source: `wc -l` and `wc -c` run directly against the file; read-at-source: direct file measurement.

GT-4 That same file carries, fully inlined, the complete procedure text for all 8 companion techniques, located at 8 “## Procedure” headings (lines 873, 957, 1011, 1057, 1125, 1196, 1262, 1304), totaling roughly 520 of the file’s 1,390 lines (≈37%) — source: `grep -n "## Procedure"` run directly against the file; read-at-source: the located line numbers and section boundaries.

GT-5 The same 8 procedures also exist as standalone reference docs under `references/` (inversion.md 174 lines, pre-mortem.md 193, trade-off.md 225, estimate.md 88, theoretical-limit.md 141, fishbone.md 141, five-whys.md 153, second-order.md 168 — 1,283 lines total) and as separate slash-command-only skill entry points under `skills/<name>/SKILL.md` (e.g. `skills/inversion/SKILL.md`, 255 lines, frontmatter `disable-model-invocation: true`) — source: `wc -l` and `head` run directly against the files; read-at-source: direct file measurement and header inspection.

GT-6 `agents/first-principles.md` (and each `skills/<name>/SKILL.md`) carries a generation header: "GENERATED — DO NOT EDIT. Source: shared/spine/SKILL-body.md [or shared/skills –write." — these files are compiled outputs of shared source files via a build script, not independently hand-maintained — source: `grep` run directly against the file headers; read-at-source: line 16 of `agents/first-principles.md` and the equivalent header in `skills/inversion/SKILL.md`.

GT-7 `references/output-template.md` (551 lines) and `references/validation-rubric.md` (611 lines) together total 1,162 lines — larger than the entire ~500-line budget by themselves — and the methodology's own text (executed by this run, per GT-2) instructs that each be opened via Read exactly once per run, on every full run, yet neither is inlined in the always-loaded agent body — source: `wc -l` plus direct quotation of the instructions in `agents/first-principles.md`; read-at-source: the quoted "Open the output template once…" and "Open the Self-Audit Gate's rubric once…" passages.

GT-8 No CI configuration, line-count/token-budget check, or lint script was found anywhere in the inspected repository — searches for `*.yml`, `*.yaml`, `*ci*`, `*budget*`, `*lint*`, and for a `.github/workflows` directory, all returned no matches — source: `find` run directly against the repository root; read-at-source: the exhaustive (over the visible tree), negative `find` result.

GT-9 The inspected repository has no accessible git history (`git log` and `git status` both fail with "not a git repository") — no commit-level evidence exists in this environment to confirm or date the scenario's stated "~500-line original budget," "two milestones," or "878-line" figures — source: `git log`, `git status` run directly; read-at-source: the commands' own negative output.

GT-10 The methodology's own instructions (Phase 5′s adversarial-technique step, executed by this run) explicitly direct the agent to open `pre-mortem.md` / `inversion.md` via a fresh Read call at the point of use even though the identical procedure text is already inlined in the same agent file, with the stated reason: "a pre-mortem run from recollection reliably drops the past-tense framing that is the technique's whole mechanism" — source: direct quotation from `agents/first-principles.md`; read-at-source: the quoted passage, located in the Phase 5 "Validate" operation text.

GT-11 Lines 348-630 of `agents/first-principles.md` (≈283 of 1,390 lines, ≈20%) are the "Deliver the analysis as a file" procedure — a multi-step shell/typst/pandoc pipeline for assembling, rendering, and indexing output artifacts (file creation, section append, figure drawing, PDF rendering, folder-index generation) — a deterministic, non-judgment job distinct from the reasoning-method content occupying the rest of the file — source: `grep -n "^# \|^## "` for section boundaries plus direct reading of the located content (already present in this run's own context per GT-1); read-at-source: lines 306-630 bounded by `grep`, with the "Deliver the analysis as a file" sub-boundary located at line 348, both confirmed against this run's own loaded instructions.

GT-12 The repository contains an executable hook, `hooks/delivery-guard.py` (422 lines), registered via `hooks/hooks.json` on the `SubagentStop` event for the `first-principles` agent, which deterministically re-checks (by reading the agent's own transcript) that the analysis file was written and assembled, and blocks the agent's stop (at most twice, fail-open on error) if not — i.e. one deterministic/mechanical concern (enforcing delivery) has already been moved out of natural-language instructions into code, while the figure/PDF/index pipeline (GT-11) has not — source: `wc -l` and direct reading of `hooks/delivery-guard.py` and `hooks/hooks.json`; read-at-source: the quoted docstring and hook registration, and the counted line length.

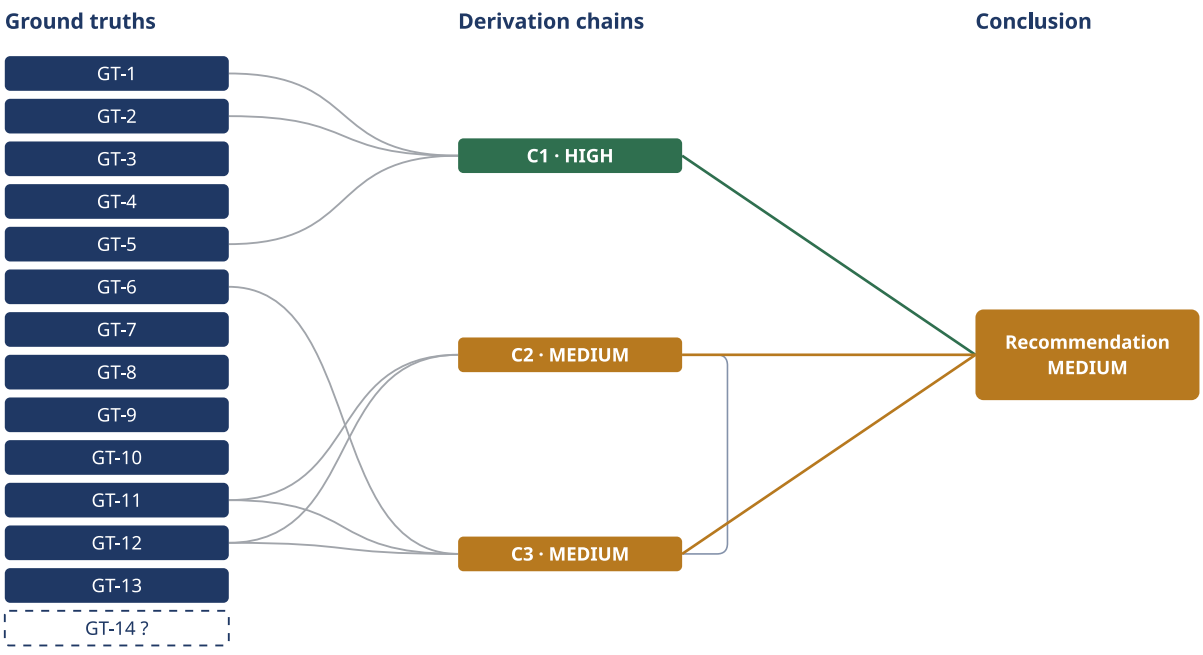
GT-13 The plugin manifest (`.claude-plugin/plugin.json`) describes the shipped product as "the 5-phase first-principles methodology plus 15 namespaced skills — 8 companion-technique skills… 5 focused-phase skills… the first-principles-analysis launcher, and the persona reader-view companion" — the maintainers'

own stated scope already spans 16 separate invocation surfaces (1 agent + 15 skills), not one file — source: direct read of `.claude-plugin/plugin.json`; read-at-source: the quoted `description` field.

GT-14? Tool-call round trips carry structural overhead beyond their raw content length (tool-name/parameter framing, and — observed directly in this run's own transcript — `<system-reminder>` blocks appended after several tool calls), so a Read call that fetches N tokens of reference content typically costs somewhat more than N tokens of equivalent inlined content once framing is counted — source: direct observation of this run's own transcript; unverified: the *qualitative* existence of nonzero overhead is directly observed here, but the general *magnitude* of that overhead across arbitrary harnesses, models, or request volumes was not measured by this run and is not read from any external source.

Provenance summary: ?-marked: GT-14 (1 of 14). Read-at-source locations for the unsuffixed ground truths feeding load-bearing chains: GT-1 (this run's own loaded instructions, companion-technique "## Procedure" sections) — chain C1; GT-2 (this run's own tool-call transcript, the two Read calls plus quoted instruction text) — chain C1; GT-3 (`wc -l` / `wc -c` on `agents/first-principles.md`) — chains C1, C3; GT-4 (`grep -n "## Procedure"` on `agents/first-principles.md`, 8 matches located) — chains C1, C3; GT-5 (`wc -l` / `head` on `references/*.md` and `skills/inversion/SKILL.md`) — chain C1; GT-6 (header line 16 of `agents/first-principles.md`) — chains C1, C3; GT-7 (`wc -l` on `references/output-template.md` and `references/validation-rubric.md`, plus the quoted Read instructions) — chain C1; GT-8 (`find` over the repo root, negative result) — chain C3; GT-9 (`git log` / `git status`, negative result) — chain C1 (as a scoping caveat); GT-10 (quoted passage, Phase 5 text) — chain C3; GT-11 (`grep` section boundaries plus direct reading, lines 348-630) — chains C2, C3; GT-12 (`wc -l`/direct reading of `hooks/delivery-guard.py` and `hooks/hooks.json`) — chains C2, C3; GT-13 (direct read of `.claude-plugin/plugin.json`) — chain C2 (context only, not load-bearing for a numeric score).

4. Derivation Chains



Solid = read at source · dashed = not read at source (?) · fill = confidence · amber dashed = unverified input

Companion technique applied in this section: Trade-off analysis (Phase 4, four options scored against six locked criteria, collapsed into chain C3 below) and Second-Order Thinking (extending C3 in place with order-marked hops, walking both the actor lens — maintainers, the agent itself, a future contributor; a competitor/adversary lens is noted as not strongly applicable to an internal plugin's budget policy — and

the time lens — immediately, after a few cycles, once assumed as convention). Estimate and Theoretical-Limit are recorded as not applicable in the Techniques-not-applied block in the appendix.

Trade-off work (full matrix, collapsed into chain C3's first two hops):

Must-haves: M1 — the option must not reduce the end-user-visible six-section output or the methodology's rigor merely to hit a number. No other must-have knockouts; all four options below satisfy M1 as stated.

Options (including status quo and a composite): O1 — split as literally proposed (move companion-tool procedures/template/rubric into separate reference files Read on demand, shrink the inline core, no change to what executes or to agent responsibility). O2 — retire or redefine the line-count budget; leave the file's structure as is. O3 — narrow scope: extract the mechanical publishing pipeline (GT-11) into a script/hook, following the precedent in GT-12; leave the budget question and companion-technique packaging alone for now. O4 (composite) — do O3 first, then re-measure the generated artifact's actual token count, then redefine the budget around that measurement, then decide any further companion-technique splitting technique-by-technique against real firing-frequency data.

Criteria, weights locked before scoring, anchors (1/5): Real cost/latency reduction (w=5; 1=always-loaded tokens unchanged or worse, 5=measurably cut in the generated artifact) · Goodhart/gaming-resistance (w=5; 1=satisfiable by moving bytes with zero change to what's loaded/executed, 5=cannot be satisfied without the underlying loaded-token or responsibility count changing) · Scope-creep addressed (w=4; 1=byte count changes only, 5=a named responsibility is relocated out of the agent's instructions or mandate entirely) · Implementation cost (w=2; 1=touches the build pipeline, every generated surface, and the hook system, 5=small, local, low-risk) · Maintainability (w=3; 1=creates files that must be hand-reconciled with no single source of truth, 5=preserves or strengthens a single generated source of truth) · Reversibility (w=2; 1=destroys the evidence needed to revisit the decision, 5=fully reversible and produces missing evidence for next time).

Scores and weighted totals (recomputed independently in the adversarial-pass record below):

Option	Cost(5)	Gaming(5)	Scope(4)	Impl(2)	Maint(3)	Rev(2)	Total
O1 split-as-proposed	2	1	1	3	2	3	37
O2 retire/redefine only	1	3	1	5	4	4	54
O3 narrow-scope-only	5	5	5	2	4	4	94
O4 composite (recommended)	5	5	5	2	5	5	99

Flip test: O4 beats O3 on every single criterion (ties on four, strictly ahead on two — Maintainability and Reversibility) and never trails on any — O3 is strictly dominated, so **no reweighting of any single criterion flips O4 vs. O3**; this is the strongest form of robustness the trade-off procedure can report. O4's lead over O2 (45 points) and O1 (62 points) is driven almost entirely by Cost, Gaming-resistance and Scope (combined weight 14 of 21) where O2/O1 score 1-3 against O4's 5s; closing that gap would require moving Implementation-cost's weight (currently 2) to roughly 10-12× its locked value relative to the other criteria — not a plausible single-criterion move, so this ranking is robust as well, though less absolutely than the O3/O4 comparison.

Conclusion C1: In this harness, whether a chunk of content is always-loaded is decided by which construct holds it, not by which physical file a maintainer put it in

GT-1 (full agent body, 8 companion techniques inlined, no Read needed) + GT-2 (template/rubric absent until an explicit Read) + GT-5 (disable-model-invocation skill frontmatter) → whether content is always-loaded in this harness is a property of the construct that holds it — a concatenated-at-build agent body, a runtime-Read reference, or a slash-command-only skill — not of which physical markdown file a maintainer happened to put it in → splitting a file reduces the always-loaded token cost only when the material moved out lands in a construct the harness does not load unconditionally, not merely when it moves to a second file

Confidence: HIGH — both hops follow by direct observation of this run's own loading behavior (GT-1, GT-2) corroborated by an independent, design-time mechanism (GT-5's `disable-model-invocation` frontmatter);

the live rival — "the harness eagerly preloads every referenced file regardless of explicit Read calls, so the construct distinction doesn't matter" — is named and ruled out in §5, Dead End 1, by GT-2 itself (this analysis could not quote the template/rubric's content before issuing the Read calls that fetched them).

Conclusion C2: A material share of this file's growth past budget is new, distinct responsibility (publishing), not deeper reasoning method

GT-11 (document-publishing pipeline, ≈20% of the file) + GT-12 (a sibling mechanical concern already relocated into a hook) → the project has already relocated one deterministic, non-judgment concern — enforcing delivery — out of natural-language instructions and into executable code, successfully → the same relocation is available for the figure/PDF/index pipeline named in GT-11, because it is equally deterministic and non-judgment *[Assumes: A-10 — prose-encoded deterministic steps are reliably portable to a script without losing the defensiveness the current prose was written to provide; no such script has been built or tested in this analysis]* → a material share of this file's growth past any line-count budget is attributable to the agent acquiring a categorically distinct responsibility — document publishing — not to the reasoning methodology itself deepening

Confidence: MEDIUM — Inputs and Rivals are clean (both head GTs are unsuffixed and read at source; the live rival that the larger 37%-of-file companion-technique content, GT-4, dominates "legitimate depth" over this 20% publishing share is not disputed — the claim above is deliberately qualified as "a material share," not "the majority," so it does not contest that rival); Inference is the short axis — the second hop rests on `[Assumes: A-10]`, and that assumption has not been priced by actually building and testing the extraction script. What would close it: build the script against the existing prose's own defensive test cases (the "NOT APPENDED"/"NOT ASSEMBLED" guards) and confirm matching behavior; until then this chain stays MEDIUM.

Conclusion C3: Narrow scope before touching either the file boundary or the budget number

GT-11 (document-publishing pipeline, ≈20% of the file) + GT-12 (sibling mechanical concern already relocated into a hook) + GT-6 (one shared source generates 16 differently-sized artifacts) + C2 (MEDIUM — growth is partly new responsibility, not just depth) → O4 (narrow scope first, then re-measure the generated artifact's token count, then redefine the budget, then decide further companion-technique splits on firing-frequency evidence) scores 99 and strictly dominates O3 (narrow-scope-only, 94) on every one of six locked criteria, an ordering that is weight-invariant → O4 also leads O2 (retire/redefine only, 54) and O1 (split-as-proposed, 37) by wide margins driven by Cost-reduction, Gaming-resistance and Scope-creep-addressed (weights 5, 5, 4) *[Assumes: A-16 — the six criteria and their locked weights capture the decision-relevant factors; the O4-over-O3 ordering is weight-invariant and not exposed to this assumption, but the O4-over-O2/O1 margins are]* → recommend O4: extract the mechanical publishing pipeline into a script/hook before, and as a precondition for, any decision to split reference files or redefine the line-count budget →[2nd] maintainers must build and test that script against the existing prose's own defensive edge cases, and a future contributor adding new agent content now faces "is this a script's job or a reasoning job" at review time instead of "does this fit under N lines" *[Assumes: A-17 — a future contributor/reviewer will actually apply that test rather than reverting to a line-count-only review]* →[2nd] the agent's own context shrinks for the mechanical section, freeing attention for the judgment-heavy phases, but the pipeline becomes less directly editable by a maintainer who can only edit markdown and cannot run code *[Assumes: A-18 — some maintainer of this file lacks the ability or willingness to edit/run a script]* →[3rd] without a dated, named deliverable for the re-measurement and redefinition steps, the plan can reproduce the original failure — an unenforced, undocumented budget — one level up, which is the pre-mortem's most damaging finding (Cluster D below) and the reason Section 6's recommendation is conditioned on a dated tripwire rather than an open-ended intention *[Assumes: A-19 — an open-ended commitment without a dated deliverable is systematically less likely to be completed than a dated one]*

Confidence: MEDIUM — Inputs axis is capped at MEDIUM because the head cites C2 (MEDIUM); C2's own confidence line carries that explanation and is not repeated here. This chain's own additional shortfall is `[Assumes: A-16]`: the O4-over-O3 dominance is weight-invariant (verified by recomputation in the adversarial-pass record below, so that part of the conclusion is not actually exposed to A-16), but the size of

O4′s lead over O2 and O1 does depend on the locked weights, and a reader who weights Implementation-cost far more heavily than this analysis does could reasonably prefer O3 alone as a faster stopgap. The 2nd/3rd-order extension steps carry their own `[Assumes: A-17]`, `[Assumes: A-18]`, `[Assumes: A-19]` but do not further lower this chain's endpoint rating, since the endpoint (recommend O4) is established by the third hop and the extension steps qualify downstream consequences rather than the recommendation itself. What would close the chain's own shortfall: have the weights set by the people who will bear the implementation cost, then re-check whether the O4-over-O3 dominance (weight-invariant, already settled) and the O4-over-O2/O1 margin (weight-dependent) both survive.

5. Abandoned Reasoning

Dead End: The harness eagerly preloads every referenced file regardless of explicit Read calls

What was tried: Before accepting GT-1/GT-2 as evidence that loading is construct-dependent, this analysis considered the rival explanation that Claude Code might eagerly preload all Read-able reference files into context regardless of whether a Read call is issued, which would make the agent-body/reference-file distinction cosmetic rather than load-bearing for chain C1.

Why abandoned: Directly contradicted by GT-2: this analysis could not quote `output-template.md`'s or `validation-rubric.md`'s content at any point before issuing the two explicit Read calls recorded earlier in this run's own transcript, and could quote both immediately after. Eager preloading would have made the content quotable from turn one; it was not.

What it ruled out: Saves a future reader of this analysis from re-litigating whether GT-1/GT-2′s "construct decides load timing" claim might be an artifact of this run's happenstance rather than the harness's actual behavior — the rival was tested directly against this run's own transcript, not merely asserted away.

Dead End: Bare budget retirement (O2) with no replacement measurement

What was tried: Considered scoring "just raise or delete the line-count budget, do nothing else" as its own first-class option distinct from O2-as-scored, on the theory that removing a broken gate is itself a sufficient fix if the gate was never well-founded.

Why abandoned: This is already exactly what O2 scores (O2 = "retire or redefine the line-count budget; leave the file's structure as is") — scored low (54) precisely because, taken bare, it resolves none of A-1/A-11 through A-15 (which goal, if any, the number served) and leaves GT-11/GT-12′s concrete scope-creep finding untouched. Giving it a separate, more favorable score would have required smuggling in the re-measurement step that properly belongs to O4, double-counting the same idea under two option labels.

What it ruled out: Saves a future reader from treating "just delete the budget" as a free action — it is cheap (Impl=5) but by itself addresses none of the higher-weighted criteria (Cost, Gaming-resistance, Scope), which is exactly why it loses to O3/O4 in the trade-off despite being the easiest option to execute.

Dead End: Uniform Read-gating of all eight companion techniques

What was tried: Considered recommending that all eight companion-technique procedures be moved out of the always-loaded agent body and into Read-gated reference files uniformly, on the theory that all eight are "auxiliary" and conditionally invoked.

Why abandoned: A-6 (discarded in the Assumptions Table): GT-4 and GT-10 show the inspected system already treats the eight techniques non-uniformly — all eight are inlined, but pre-mortem and inversion-as-adversary are additionally forced through a fresh Read at their point of use, for a stated fidelity reason (GT-10), not a frequency reason. Recommending uniform treatment would discard evidence already present in the codebase about how these specific techniques actually need to be invoked, and would not obviously change the always-loaded cost by much, since GT-4′s inlined content (≈37% of the file) is dominated by exactly the techniques (fishbone, second-order, estimate, theoretical-limit) for which no firing-frequency data exists to justify Read-gating one way or the other.

What it ruled out: Saves a future reader from treating “the six/eight companion tools” as one bucket for a splitting decision — this analysis’s recommendation (O4) explicitly defers that decision to per-technique frequency evidence rather than resolving it uniformly here, because the evidence to resolve it uniformly does not exist in what was inspected.

6. Conclusion

Recommended approach: Do not split the file as the first move. Narrow scope first: extract the deterministic “Deliver the analysis as a file” publishing pipeline (figure/PDF/index rendering) into an executable script or hook, following the project’s own precedent for a sibling concern (chain C2; chain C3). Then re-measure the generated agent artifact’s actual token count, redefine the line-count budget around that measurement (not around the hand-maintained source file’s line count), and only then decide, technique-by-technique and on real firing-frequency evidence, whether any of the eight companion-technique procedures are worth Read-gating out of the always-loaded body (chain C3).

Key insight: The reflexive “split the file” move conflates two independent questions — where instructions physically live, and what the agent is responsible for doing — and in the one real instance inspected, roughly a fifth of the file’s length is not reasoning-method content at all but a second, distinct job (document publishing) the agent quietly acquired; that responsibility, not file organization, is the part worth relocating first (chain C1; chain C2).

Trade-offs acknowledged: Extracting the publishing pipeline into a script makes it less directly inspectable and editable by a maintainer who only edits markdown and cannot run code, and the recommended re-measurement/redefinition steps add real, if modest, implementation delay before any budget question is resolved (chain C3). A further risk, surfaced by this analysis’s own pre-mortem (Cluster D, in the adversarial-pass record below): an open-ended “we’ll measure it later” commitment can reproduce the original undocumented-budget failure one level up unless it is given a dated, named deliverable — no chain — flagged assumption only.

Confidence: MEDIUM — matches C3’s own band, the weakest chain this Conclusion rests on; C2 and C3 each already name their own downgrade cause (`[Assumes: A-10]` on C2; `[Assumes: A-16]` on C3) and need not be re-explained here. The Conclusion’s own open question, not fully closed by any chain: whether the original ~500-line budget was ever derived from a measured target, and whether a CI gate matching the scenario currently exists and is being satisfied or silently bypassed, cannot be settled from the evidence available (GT-8, GT-9) — this analysis recommends the dated-measurement step in O4 precisely because that evidence does not yet exist, not because this analysis has it.